

Three AI systems agreed with each other. Not with us.

AI is increasingly being asked to make the same judgment call you make every session: figuring out why a student got something wrong. Before it shapes how your students are taught, we checked whether it can be trusted — and wanted to walk you through what we found.

Research briefing • **100** real tutoring conversations • **3** AI systems • **2** expert human reviewers

BACKGROUND

What is "learning analytics," and why do we care?

IN PLAIN TERMS

Studying data about how students learn — including tutoring conversations — to see where they're stuck, and why. Tutors do this by ear already. AI can now do it at a scale no person could.

WHY IT MATTERS HERE

Done well, that gets students the right help, faster. Done poorly — if the AI misreads the conversation — it points them toward the wrong help instead.

This briefing tests one question: **is the AI reading those conversations correctly?**

One wrong answer, many possible reasons

WHAT TUTORS ALREADY KNOW

- A wrong answer rarely means just one thing.
- It could be a guess, a mix-up, the wrong method, a shaky concept, or a small slip.
- Different reasons need different responses.

TODAY'S QUESTION

- We asked 3 AI systems to diagnose 100 real wrong answers.
- We checked their answers against our own expert reviewers.
- If the AI systems agree with each other, are they right?

Not necessarily. Here's what we found.

Six reasons a response can be wrong

LABEL	WHAT IT MEANS	WHAT IT SOUNDS LIKE
UNC	Uncertainty — didn't really commit to an answer	<i>"Um, is it maybe 40?"</i>
CTX	Misattribution — reasoning imported from a different problem	<i>Answers using a number from question 4, not question 5</i>
COMP	Operator selection — the wrong operation or approach for this question	<i>Adds when the problem calls for multiplying</i>
CON	Conceptual gap — a wrong underlying idea	<i>Believes multiplying by 10 means "add a zero," even to a decimal</i>
PROC	Procedural slip — right approach, execution mistake	<i>Correct method, arithmetic error partway through</i>
N/A	Not diagnosable — can't be determined from the transcript	<i>Inaudible, off-task, or too tangled to pin on one cause</i>

Every wrong answer got exactly one label. "Not diagnosable" isn't a sixth kind of mistake — it's an honest "we can't say."

A fair test, not a demo

100 real wrong answers. 3 AI systems diagnosed why — seeing only what a tutor would see in the moment, not the correction after. Two expert reviewers did the same, then agreed on one answer per item to set the standard.

100

real tutoring conversations checked

3

AI systems compared, one per major vendor

2

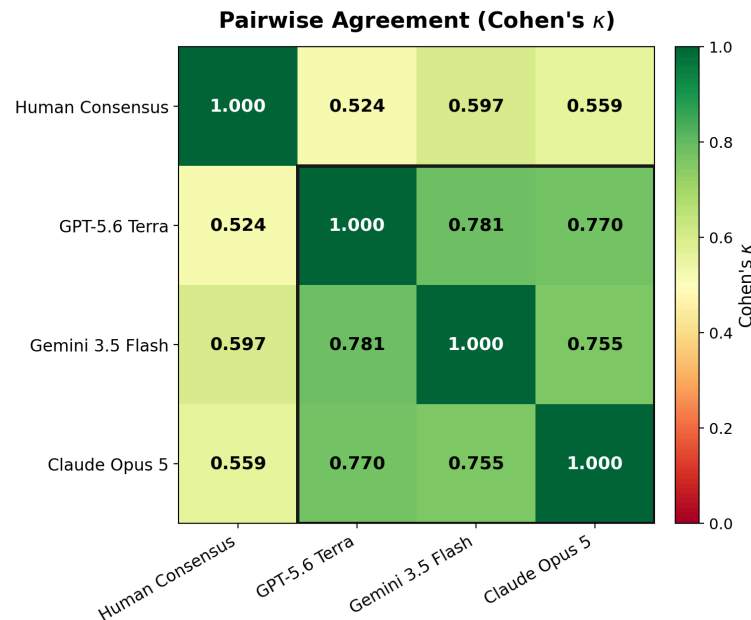
expert human reviewers, reconciled to one answer

OpenAI · GPT-5.6 Terra

Google · Gemini 3.5 Flash

Anthropic · Claude Opus 5

The AI systems agreed with each other far more than they agreed with us



How to read this: each square scores how closely two raters agreed, 0 to 1 — greener means closer agreement. The outlined block, bottom right: the 3 AI systems agree with *each other* (.76–.78). The top row and left column: each AI system agrees with our reviewers less (.52–.60). Same systems, same data — just a different partner to agree with.

WHAT ELSE WE FOUND

The AI systems weren't just wrong — they were wrong the same way

- A small execution slip got mistaken for a deep conceptual gap **6 times out of 10** — the sharpest of these mix-ups.
- When unsure, the AI systems defaulted to the same guess: "**conceptual gap.**"
- Our reviewers said "**I can't tell**" 1 in 5 times. The AI systems said it about half as often — they'd rather guess.
- A stronger, slower AI didn't fix this. Results came out essentially the same.

QUESTIONS WELCOME

Let's talk about what comes next

Not to scrap AI-assisted diagnosis — to use it in a way that earns your trust. Tell us what matches what you're seeing, and where human review matters most.

A question for you: Do these particular model errors matter instructionally — and if so, how? For example, does mistaking a procedural slip for a conceptual gap actually change what help a student would get?

A follow-up: What's the risk of providing the wrong type of feedback to a student?